

Factor Maps

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Definition 0.1 (Kra et al., 2022, Definition 2.1). For a system (X, μ, T) we say that the system (Y, ν, S) is a *measurable factor* of (X, μ, T) if there is a measurable map $\pi : X \rightarrow Y$, the *measurable factor map*, such that $\pi(\mu) = \nu$ and $(S \circ \pi)(x) = (\pi \circ T)(x)$ for μ -almost every $x \in X$.

Definition 0.2 (Jamneshan and Kreidler, 2025, Lemma 8.11). Let $(X, \mathcal{B}), (Y, \mathcal{C})$ be probability spaces and let $U : L^2(X) \rightarrow L^2(Y)$ be a Markov embedding. The *conditional expectation* of $f \in L^2(X)$ on (Y) is the orthogonal projection of f on $U(L^2(Y))$, denoted $\mathbb{E}_\mu(f | Y)$, such that

1. (Linearity) $f \mapsto \mathbb{E}_\mu(f | Y)$ is a linear operator from $L^2(X)$ to $L^2(Y)$, i.e.,

$$\mathbb{E}_\mu(f + g | \mathcal{C}) = \mathbb{E}_\mu(f | \mathcal{C}) + \mathbb{E}_\mu(g | \mathcal{C}).$$

2. (Non-Negative) If $f \geq 0$, then $\mathbb{E}_\mu(f | Y) \geq 0$.
3. If $f \in L^2(Y)$, then $\mathbb{E}_\mu(U(f) | Y) = f$. In particular, $\mathbb{E}_\mu(1 | Y) = 1$.
4. If $f \in L^2(X)$ and $g \in L^\infty(Y)$, then $\mathbb{E}_\mu(fU(g) | Y) = g\mathbb{E}_\mu(f | Y)$.
5. For $f \in L^2(X)$,

$$\int_X f \, d\mu_X = \int_Y \mathbb{E}_\mu(f | Y) \, d\mu_Y.$$

6. For $f \in L^2(X)$, the conditional expectation $\mathbb{E}_\mu(f | Y)$ is the unique element of $L^2(Y)$ satisfying

$$\int_Y \mathbb{E}_\mu(f | Y) h \, d\mu_Y = \int_X f(U(h)) \, d\mu_X,$$

for all $h \in L^\infty(Y)$.

7. For $f \in L^2(X)$,

$$|\mathbb{E}_\mu(f | Y)|^2 \leq \mathbb{E}_\mu(|f|^2 | Y).$$

8. The conditional expectation operator extends to a linear operator from $L^1(X)$ to $L^1(Y)$ satisfying the properties (1)-(5). Moreover, it maps each $L^p(X)$ to $L^p(Y)$, for $1 \leq p \leq \infty$, with $\|\mathbb{E}_\mu(f | Y)\|_{L^p(Y)} \leq \|f\|_{L^p(X)}$ for every $f \in L^p(X)$.
9. (Monotone Convergence) If $0 \leq f_1 \leq f_2 \leq \dots$ is a monotone sequence and f is an element in $L^2(X)$ such that $(f_n)_{n \in \mathbb{N}}$ converges to f almost everywhere, then $\mathbb{E}_\mu(f_n | Y)$ converges to $\mathbb{E}_\mu(f | Y)$ almost everywhere.

10. (Dominated Convergence) If $(f_n)_{n \in \mathbb{N}}$ is a sequence and f is an element in $L^2(X)$ such that f_n converges to f almost everywhere and there is $g \in L^1(X)$ with $|f_n| \leq g$ almost everywhere for all n , then $\mathbb{E}_\mu(f_n | Y)$ converges to $\mathbb{E}_\mu(f | Y)$ almost surely.

In the case that $Y = X$, then we instead denote the conditional expectation as $\mathbb{E}_\mu(f | \mathcal{G})$.

Whilst looking at conditional expectations and understanding property (5), we remarked that $\mathbb{E}_\mu(f | \mathcal{G})$ averages a function $f : X \rightarrow \mathbb{C}$ in $(X, \mathcal{B})$ on each subset in $\mathcal{G}$.

Definition 0.3 (Kra et al., 2022, sect. 2). Fix a system (X, μ, Γ) and a probability space (Ω, ν) . Whenever we have a map $\omega \mapsto \mu_\omega$ from Ω to $\mathcal{M}(X)$, the set of probability measures on X , with the properties

1. $\omega \mapsto \mu_\omega(B)$ is measurable for every Borel set $B \subset X$,
2. $\int_X f \, d\mu = \int_\Omega \int_X f \, d\mu_\omega \, d\nu(\omega)$ for all measurable and bounded $f : X \rightarrow \mathbb{C}$,

we call $\omega \mapsto \mu_\omega$ a *disintegration* of μ .

Theorem 0.1. If $\pi : (X, \mathcal{B}, \mu) \rightarrow (Y, \mathcal{G}, \nu)$ is a measurable factor map, then there exists a disintegration $y \mapsto \mu_y$ of μ defined on (Y, ν) such that

$$\mathbb{E}_\mu(f | Y)(y) = \int_X f \, d\mu_y$$

for ν -almost every $y \in Y$, whenever $f : X \rightarrow \mathbb{C}$ is measurable and bounded. Furthermore, if $y \mapsto \eta_y$ is another such disintegration, then $\eta_y = \mu_y$ for ν -almost every $y \in Y$.

Definition 0.4 (Kra et al., 2022, Definition 2.7). Let (X, μ, Γ) be a system and $\mathcal{G}$ be a σ -algebra of G -invariant Borel sets. An *ergodic decomposition* of μ is a disintegration $x \mapsto \mu_x$ of μ defined on (X, μ) such that

$$\int_X f \, d\mu_x = E(f | \mathcal{G})(x)$$

holds for μ -almost every $x \in X$, whenever $f : X \rightarrow \mathbb{C}$ is measurable and bounded.

Notice that, by Theorem 0.1, we immediately have the property that, if $x \mapsto \nu_x$ is another ergodic decomposition of (X, μ, Γ) , $\mu_x = \nu_x$ for μ -almost every $x \in X$.

In working on Exercise 8.5 provided by Jamneshan and Kreidler (2025), we also reach the following result:

Theorem 0.2 (cf. Jamneshan and Kreidler, 2025, Exercise 8.5). For a system (X, μ, Γ) , there exists an ergodic decomposition $x \mapsto \mu_x$ where, for μ -almost every $x \in X$, the measure μ_x is Γ -invariant and ergodic.

This is proved by starting with Definition 0.3 and Theorem 0.1 and using the assumptions that $Y = X$, $\nu = \mu$, and $\mathcal{G}$ is the σ -algebra of Γ -invariant Borel

sets. We then use the fact that $L^2(Y, \mathcal{C}) \subseteq L^2(X, \mathcal{B})$ to conclude that there exists a Markov embedding

$$U : L^2(Y, \Sigma_Y, \mu_Y) \rightarrow L^2(X, \text{Bor}(X), \mu_X)$$

where $U(L^2(Y, \Sigma_Y, \nu)) = L^2(X, \Sigma_Y, \mu)$. We then show that $(X, \mathcal{B}, \mu_y, \Gamma)$ is a measure-preserving system by showing that $(U_\gamma)_* \mu_y = \mu_y$ for every $\gamma \in \Gamma$ and almost every $y \in Y$ using Theorem 0.1. We conclude the proof by proving ergodicity by arriving to the conclusion that $\mu_y(E) = \mathbb{1}_E(y) \in \{0, 1\}$ for some $E \in \mathcal{C}$.

Jamneshan, A. and Kreidler, H. (2025). *ISem 28: Ergodic structure theory and applications*.

Kra, B. et al. (2022). *Infinite sumsets in sets with positive density*. Available at: <https://arxiv.org/abs/2206.01786>